

WEEK 3 – DATA SCIENCE AND VISUALIZATION

The Cleaning Sprint: Missing Value Handling and Outlier Detection

Project: Spotify Songs Analysis

1. Introduction

Week 3 focuses on data cleaning and quality assessment. Missing values and unusual observations can affect statistical calculations, visualizations, and subsequent analysis. The main tasks were to identify missing values, apply suitable imputation strategies, detect potential outliers using boxplots and Z-score analysis, and compare the dataset before and after cleaning.

2. Objectives

The objectives were to identify missing values, analyze their distribution, apply appropriate numerical and categorical imputation, handle columns with excessive missing data, detect potential outliers using boxplots and Z-score, quantify detected outliers, and verify the final cleaned dataset.

3. Dataset Used

The analysis used the Spotify Songs dataset containing approximately **19,335 song records**. Important numerical audio features included danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, duration, and time signature.

4. Tools and Technologies Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, and SciPy were used for data cleaning, statistical analysis, and visualization.

5. Missing Value Analysis – Before Cleaning

The first step was to identify missing values across all columns. The analysis showed that several numerical audio features contained **7 missing values each**. The most significant missing-value problem was found in **preview_url**, with **7,524 missing values**. Several metadata fields contained smaller numbers of missing values.

```
df.isnull().sum()
```

Observation: Missing values were not uniformly distributed. Therefore, different treatment strategies were required rather than applying one method to every column.

6. Missing Value Handling

Different strategies were selected according to the type and purpose of each column.

6.1 Median Imputation

Median imputation was applied to numerical audio features. The median is less affected by extreme values than the mean, making it appropriate for features that may contain skewness or potential outliers.

```
for col in numeric_cols:
    if df[col].isnull().sum() > 0:
        df[col] = df[col].fillna(df[col].median())
```

6.2 Mode Imputation

Mode imputation was applied to the categorical column **album.type**. The mode represents the most frequently occurring category.

```
df["album.type"] = df["album.type"].fillna(
    df["album.type"].mode()[0]
)
```

6.3 Handling preview_url

The **preview_url** column contained 7,524 missing values. Since it is metadata rather than an analytical audio feature and a large proportion of its values were missing, the column was removed instead of being artificially imputed.

```
df = df.drop(columns=["preview_url"])
```

7. Missing Value Analysis – After Cleaning

After imputation, the selected numerical audio features were checked again. Each of the previously affected numerical features had zero remaining missing values.

```
df[numeric_cols].isnull().sum()
```

Before: 7 missing values per affected numerical feature.

After: 0 missing values in the selected numerical features.

8. Outlier Detection Using Boxplots

Boxplots were used to visually identify observations lying outside the whiskers. Potential outliers were observed in several features, particularly loudness, speechiness, instrumentalness, liveness, danceability, and tempo.

```
plt.figure(figsize=(8, 5))
sns.boxplot(x=df["danceability"])
plt.title("Boxplot of Danceability")
plt.xlabel("Danceability")
plt.show()
```

Interpretation: Values outside the whiskers are potential outliers, but they are not automatically errors. In a music dataset, unusual audio characteristics can represent legitimate songs.

9. Outlier Detection Using Z-score

Z-score analysis was used to quantify how far observations were from the mean. An absolute Z-score greater than 3 was treated as a potential outlier.

```
from scipy.stats import zscore

z_scores = df[numeric_features].apply(zscore)
outlier_counts = (abs(z_scores) > 3).sum()
```

The potential outlier counts and percentages were:

Feature	Potential Outliers	Percentage
Danceability	90	0.47%
Energy	0	0.00%
Loudness	479	2.48%
Speechiness	336	1.74%
Acousticness	0	0.00%
Instrumentalness	886	4.58%
Liveness	524	2.71%
Valence	0	0.00%
Tempo	53	0.27%

10. Outlier Analysis – Before and After

Before quantitative analysis, boxplots indicated potential extreme observations. Z-score analysis then provided a numerical measure of these observations. Instrumentalness had the highest potential-outlier percentage at **4.58%**, followed by liveness at **2.71%** and loudness at **2.48%**. Energy, acousticness, and valence had no observations beyond the selected Z-score threshold.

The detected outliers were **retained** rather than removed. This was because extreme audio-feature values can represent genuine characteristics of songs and are not necessarily data-entry errors. Therefore, the outlier analysis was used for detection and interpretation rather than automatic deletion.

11. Descriptive Statistics After Cleaning

Descriptive statistics were calculated to understand the central tendency and spread of the cleaned audio features.

```
df[numeric_features].describe().T
```

Important averages included danceability $\approx$ 0.616, energy $\approx$ 0.584, acousticness $\approx$ 0.309, valence $\approx$ 0.491, tempo $\approx$ 119.79 BPM, and loudness $\approx$ -8.60 dB.

12. Final Cleaning Verification

The selected numerical features were checked again to confirm successful missing-value treatment.

```
df[numeric_features].isnull().sum()
```

Result: zero missing values remained in the selected numerical audio features.

12.1 Saving the Cleaned Dataset

```
df.to_csv("spotify_cleaned_week3.csv", index=False)
```

The cleaned dataset was saved for use in the subsequent exploratory data analysis stage.

13. Summary of Cleaning Process

Data Quality Issue	Method Applied	Result
Missing numerical values	Median imputation	Successfully handled
Missing categorical value	Mode imputation	Successfully handled
Excessive missing metadata	Column removal	preview_url removed
Potential outliers	Boxplots	Visually identified
Potential outliers	Z-score	Quantitatively identified
Outlier removal	Not performed	Values retained as potentially legitimate

14. Conclusion

Week 3 improved the quality and reliability of the Spotify Songs dataset through systematic data cleaning. Missing values were identified and treated using suitable strategies: median imputation for numerical features and mode imputation for categorical data. The preview_url column was removed because of its high proportion of missing values and limited analytical importance. Outliers were investigated using boxplots and Z-score analysis. Potential outliers were retained because they may represent legitimate characteristics of songs rather than errors. Overall, the selected analytical audio features contained no missing values after cleaning, and the cleaned dataset was saved for the next stage of the project: Exploratory Data Analysis (EDA).

15. Project Status

Week 3: Handling Missing Values and Outlier Detection — Completed